

TUKTOYAKTUK, NT, Canada

WMO: 719850

Lat: 69.436N

Lon: 133.023W

Elev: 18

StdP: 14.69

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-37.5	-34.7	-45.1	0.4	-36.8	-42.1	0.5	-34.4	41.1	-0.7	32.5	3.5	3.8	90	0.973

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.8	72.7	60.0	68.5	58.2	64.1	56.3	61.6	69.9	59.1	66.7	56.8	63.3	9.7	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.6	70.9	65.4	55.4	65.4	62.6	53.2	60.3	60.4	27.5	69.5	25.9	66.9	24.4	63.4	70.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.3	23.1	20.4	DB	-41.2	79.7	3.5	3.9	-43.7	82.4	-45.8	84.7	-47.8	86.9	-50.3	89.7	(4)
(5)			WB	-41.4	65.1	3.5	2.8	-43.9	67.1	-45.9	68.7	-47.9	70.3	-50.4	72.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.2	-13.8	-13.7	-11.3	5.7	27.3	44.6	52.7	48.7	38.8	21.4	1.9	-9.8
	DBStd	26.91	12.71	12.56	11.86	12.06	11.02	8.54	8.29	7.57	6.88	9.02	10.52	11.61
	HDD50	12619	1977	1783	1901	1328	707	211	68	118	343	888	1442	1853
	HDD65	17818	2442	2203	2366	1778	1170	613	388	508	787	1353	1892	2318
	CDD50	285	0	0	0	0	2	48	152	77	6	0	0	0
	CDD65	10	0	0	0	0	0	1	7	2	0	0	0	0
	CDH74	69	0	0	0	0	0	10	49	10	0	0	0	0
	CDH80	5	0	0	0	0	0	0	4	1	0	0	0	0
Wind	WSAvg	9.9	9.6	9.8	9.0	10.0	10.3	10.3	10.5	10.4	10.5	9.8	9.4	9.0
Precipitation	PrecAvg	5.4	0.2	0.2	0.1	0.2	0.2	0.5	0.9	1.2	0.7	0.6	0.3	0.3
	PrecMax	8.3	0.7	0.7	0.6	0.7	1.3	1.8	1.9	3.4	1.5	1.4	0.9	0.7
	PrecMin	2.7	0.0	0.0	0.0	0.0	0.0	0.0	0.3	0.4	0.0	0.1	0.0	0.0
	PrecStd	1.3	0.2	0.2	0.2	0.2	0.3	0.4	0.5	0.7	0.4	0.4	0.2	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.7	23.4	24.6	37.8	61.5	74.6	78.6	74.4	62.9	41.6	30.9	22.1
		MCWB	26.1	22.4	23.4	34.9	49.6	58.7	61.6	62.8	53.5	39.3	29.8	21.4
	2%	DB	18.3	15.7	17.0	33.7	52.6	69.5	74.6	68.3	56.1	36.3	24.6	15.7
		MCWB	17.7	14.9	16.0	31.5	44.5	56.5	61.0	59.2	49.5	34.8	23.8	15.2
	5%	DB	10.0	10.5	11.2	29.2	45.8	64.4	70.7	63.7	51.4	34.0	20.6	11.2
		MCWB	9.5	9.8	10.4	27.6	40.6	54.3	59.7	57.0	47.6	32.8	19.8	10.6
	10%	DB	3.5	5.4	5.8	24.3	40.7	59.2	66.3	59.8	48.0	32.0	16.5	6.9
		MCWB	2.9	4.8	5.0	23.1	37.0	51.5	58.2	54.9	45.3	30.9	15.7	6.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.8	22.6	23.4	35.5	50.2	61.1	65.3	63.9	54.0	39.2	29.7	21.7
		MCDB	26.4	23.5	24.4	37.6	60.7	71.6	74.8	73.1	61.7	41.1	30.5	22.1
	2%	WB	17.6	15.0	15.8	31.6	45.0	57.4	62.6	60.2	50.4	35.1	23.8	15.2
		MCDB	18.2	15.7	16.6	33.5	52.6	67.2	71.0	66.4	54.7	36.0	24.5	15.7
	5%	WB	9.5	9.8	10.4	27.7	40.7	54.8	60.4	57.6	47.8	32.9	19.8	10.7
		MCDB	10.0	10.5	11.1	29.2	45.8	64.0	68.7	62.6	51.0	33.9	20.6	11.3
	10%	WB	2.9	4.7	5.1	23.0	37.1	51.5	58.3	55.3	45.4	30.9	15.8	6.4
		MCDB	3.5	5.2	5.9	24.3	40.3	59.1	65.7	59.2	48.0	31.8	16.5	6.9
Mean Daily Temperature Range	5% DB	MDBR	12.1	12.6	14.0	15.4	12.6	15.7	13.8	11.0	8.6	8.3	11.4	11.4
		MCDBR	18.5	17.2	16.6	15.9	21.3	25.4	21.2	17.5	15.1	7.1	12.6	16.2
		MCWBR	18.4	17.1	16.2	14.7	14.3	14.8	10.4	9.8	9.8	6.3	12.5	16.1
	5% WB	MCDBR	18.6	17.1	16.6	15.8	20.8	24.0	19.2	16.2	14.1	7.0	12.6	16.0
		MCWBR	18.6	17.0	16.2	14.6	14.0	14.7	10.8	9.9	9.5	6.3	12.5	16.0
Clear-Sky Solar Irradiance	taub	0.157	0.208	0.248	0.304	0.342	0.336	0.352	0.336	0.296	0.227	0.157	N/A	
	taud	1.900	2.070	2.200	2.125	2.192	2.400	2.455	2.519	2.551	2.192	1.911	N/A	
	Ebn at Noon	15	179	249	262	265	272	263	255	237	167	10	0	
	Edh at Noon	1	13	23	34	36	31	29	24	17	11	1	0	
All-Sky Solar Radiation	RadAvg	5	131	646	1369	1935	2037	1627	1021	529	163	18	0	
	RadStd	1	13	29	65	108	153	138	98	49	16	2	0	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	+1.78	+2.54	+3.15	N/A	N/A	N/A	-619	-644	N/A	N/A
Regional (8 neighbors)	+1.87	+3.26	+3.69	N/A	N/A	N/A	-769	-760	N/A	N/A

Nomenclature: See separate page